

Session 2 (FIND & READ): Solutions and facilitation notes

Foundations of Clinical Research · Level 1 · Neudata · *Instructor copy. Do not hand out before the exercises.*

These are **model** answers. Many other answers are acceptable: reward work that is specific, consistent and could be acted on. Use the "what good looks like" boxes when you circulate.

Recap quiz: Session 1 answers

1. **Audit**: checking practice against an agreed standard.
2. **P** ICU staff · **I** hand-hygiene feedback · **C** no feedback · **O** hand-hygiene compliance · **T** 3 months.
3. **Feasible**: a mortality outcome needs more than 1,000 patients.
4. **"To estimate the proportion with delirium"**: a measurable verb. "Understand" is not measurable.
5. **H0**: ICU length of stay is the same with early mobilisation and with usual care.

If many miss 4 or 5, spend one extra minute: objectives use measurable verbs; H0 always states "no difference".

Check your understanding (segment 2.1): AND, OR or NOT?

1. **AND**. 2. **OR**. 3. **Fewer** (AND narrows). 4. It removes every paper that mentions children anywhere, including good adult studies that mention children once.

Check your understanding (segment 2.2): OK or not OK?

1. **Not OK**: patchwriting is plagiarism even with a citation. 2. **OK** (but rare in medical writing). 3. **Not OK**: someone else's finding needs a citation. 4. **Not OK**: you may repeat an error, so read and cite the original.

Check your understanding (segment 2.3): spot the spin

1. Causal word for an observational design. Rewrite as *"was associated with lower pneumonia"*.
 2. $p = 0.31$ is no evidence of a difference. Rewrite as *"no significant difference; the confidence interval was wide"*.
 3. Headline is a secondary outcome. Lead with the primary result instead.
 4. Before–after change without a comparison group. Rewrite as *"mortality was lower after implementation; without a control group we cannot attribute this to the protocol"*.
-

Exercise 1: Boolean search builder

Warm-up: septic shock OR fluids gives more hits (either term). sepsis gives more hits than sepsis NOT neonatal (NOT removes papers).

Model string for the running example (as in the demo):

```
("Shock, Septic"[Mesh] OR "septic shock"[tiab] OR sepsis[tiab])
AND ("Fluid Therapy"[Mesh] OR "fluid restrict*"[tiab] OR "restrictive fluid*"[tiab] OR "conservative
AND ("Mortality"[Mesh] OR mortality[tiab] OR death[tiab])
```

Sensible extra synonyms for the I block: de-resuscitation[tiab], deresuscitation[tiab], "fluid balance"[tiab], "Water-Electrolyte Balance"[Mesh].

Sensible filters (each with a reason): article type *Randomized Controlled Trial + Systematic Review* (treatment question); *Adult: 19+ years* (our population); publication date *10 years* (current practice). Avoid *Free full text* (loses key papers).

Model string for Scenario C (delirium)

```
(Delirium[Mesh] OR delirium[tiab] OR "acute confusion"[tiab])
AND ("Hip Fractures"[Mesh] OR "hip fracture*"[tiab])
AND (prevent*[tiab] OR "multicomponent"[tiab] OR "non-pharmacological"[tiab] OR bundle[tiab])
```

Common faults

- Missing brackets: A OR B AND C is read differently from (A OR B) AND C.
- AND inside a block (septic shock AND sepsis). Synonyms need OR.
- A block for the comparison ("usual care"). It adds nothing and loses papers.
- NOT used to remove a topic the paper only mentions in passing.
- Search words in the local language: PubMed indexes English titles and abstracts.

Exercise 2: Paraphrasing (model answers)

Part A

Version	Verdict	Why
a)	Problem: patchwriting	Same structure and almost the same words as the source; one word swapped ("mortality" for "death"). Plagiarism even though it is cited.
b)	OK	New structure, adds context (matched registry cohort, 28-day mortality), keeps the calibrated word "associated", gives numbers and the citation.
c)	Problem: distorts the source	"Giving more fluid ... doubles the risk" turns an observational <i>association</i> into causation, and "doubles" overstates (HR 2.13 only for day 2; 1.56 for day 3). Citation does not fix misrepresentation.
d)	Problem: no citation	Good wording, but presenting someone else's finding without a citation is plagiarism.

Part B: model paraphrase

"Mortality and days alive without organ support at day 28 did not differ between the two groups, although the pilot trial was too small to detect such differences [1]."

What good looks like: new structure; facts unchanged (28 days, mortality, days alive without life support, similar); no stronger claim (do **not** write "restriction is safe"); citation. Adding "the trial was too small" is accurate context from the paper (48 patients; not powered for mortality).

Exercise 3: Paper 1's abstract (answer key)

Item	Answer (from the abstract)
Design (and blinding)	Multicentre, randomised, controlled pilot trial; single-blind (patients and statisticians blinded, clinicians not)
P	Adults 18–85 years with septic shock, in the first 24 h of vasopressor infusion, in three ICUs; excluded: AKI needing renal replacement, end-stage CKD, severe malnutrition
I vs C	Optimised fluid restriction (reducing maintenance fluids and fluids for drug dilution during the first 7 days) vs standard fluid strategy
Primary outcome	Cumulative fluid balance (ml/kg) in the first 5 days , not mortality
Numbers	1,201 screened; 50 randomised; 2 withdrew (control); 48 analysed (24 per group)
Main result	Median cumulative balance at day 5: 6.9 vs 35.0 ml/kg; absolute difference 13.2 ml/kg (95% CI –15.2 to 41.6), $p = 0.42$
Conclusion	"...did not reduce the overall fluid balance at day 5"
Matches the result?	Yes. The CI includes zero, and the conclusion says "did not reduce".
Spin in the abstract?	No in the abstract conclusion. (Spin appears in the <i>Discussion</i> , which calls $p = 0.42$ a "non-significant trend".) Watch-point: the fluid <i>volumes</i> (89.7 vs 114.3 ml/kg, $p = 0.0506$) are a secondary result, and the abstract reports four decimals for a borderline p . This is mild emphasis, not a claim.
Registration	ClinicalTrials.gov NCT04947904 (registered 1 July 2021, before recruitment began in September 2021)
Answers our 28-day mortality question?	No. It is a small pilot powered for fluid balance. 28-day mortality was a secondary outcome and was similar between groups: "no evidence of a difference", not "no difference".

Fast finishers (Paper 2): the title says "**Impact**" (causal); the abstract conclusion correctly says "**associated**".

Common errors to correct in the debrief

- Writing "mortality" as the primary outcome.
- Copying only the p-value and not the estimate with its 95% CI.

- Calling the trial "double-blind". Clinicians, who prescribe the fluids, were not blinded.
- Reading the medians (6.9 vs 35.0) as "five times less fluid balance, so it works". The CI crosses zero.

Exercise 4: Proposal builder 2 (what good looks like)

Continuing the worked model from Session 1 (the feasible rewrite of the running example).

Search (documented): PubMed, date, the string from Exercise 1, filters (article type RCT + systematic review; adults; 10 years), number of hits; plus a check of ClinicalTrials.gov for ongoing trials.

Working title: *Restrictive versus usual fluids in adults with septic shock: a pilot randomised trial* (13 words).

Keywords: Fluid Therapy; Water-Electrolyte Balance; Critical Care; Vasoconstrictor Agents (MeSH terms not already in the title).

Problem statement (≈ 300 words, three paragraphs)

1. *Known:* septic shock is a common cause of ICU death. Early fluid resuscitation restores perfusion, but continued fluid often produces a positive fluid balance, which observational studies associate with higher mortality (e.g. Hyun 2023).
2. *Not known:* large trials of restrictive fluids (CLASSIC 2022; CLOVERS 2023) found no mortality difference, and a pilot targeting hidden "non-resuscitative" fluids (Boulet 2024) did not significantly reduce fluid balance at day 5. Whether a restrictive protocol can reduce fluid balance in *our* setting is unknown.
3. *Why here:* a 3-month audit in our ICU found that most septic shock patients were fluid-positive by day 3 (state the number from your own audit), and practice varies between consultants.

Five key references (model)

#	Reference	Why included
1	Boulet N, et al. <i>Crit Care</i> 2024;28:429 (pilot RCT, n = 48)	Closest design and outcome; primary outcome not significantly different
2	Hyun DG, et al. <i>Ann Intensive Care</i> 2023;13:105 (matched cohort)	Positive balance on days 2–3 associated with higher 28-day mortality
3	Meyhoff TS, et al. (CLASSIC) <i>N Engl J Med</i> 2022;386:2459–70	Large RCT, septic shock, no difference in 90-day mortality
4	Shapiro NI, et al. (CLOVERS) <i>N Engl J Med</i> 2023;388:499–510	Large RCT, sepsis-induced hypotension, restrictive vs liberal, no difference in 90-day death before discharge
5	Evans L, et al. Surviving Sepsis Campaign guidelines 2021. <i>Crit Care Med</i> 2021;49:e1063–e1143	Current guideline; shows the remaining uncertainty

Instructor note: check citation details against PubMed before distributing. Groups must verify every reference themselves, which is exactly the habit to practise.

Marking guide for circulating (not a grade)

Element	Good	Needs work
Search	String, database, date, filters and hits recorded; brackets correct	"We googled it"; no date; AND inside a block
References	5, relevant, include the best available design, exported to a reference manager	Websites, textbooks only, typed by hand, unread
Literature matrix	Same columns for every paper; limitation column filled	Copied abstract sentences
Problem statement	Known → gap → local reason, each fact cited, ends with the question	A literature dump, or no local reason
Title & keywords	10–16 words; P, I/E, C and design visible; calibrated words; keywords add MeSH not in the title	"Fluids in the ICU"; abbreviations; "impact of" for an observational design
Integrity	Own words, every claim cited	Sentences pasted from abstracts

Exit question: answers

1. **OR** gives more hits.
2. Paper 1's primary outcome was **cumulative fluid balance at day 5 (ml/kg)**, not mortality.